Python Activity-Based Learning
Program 2: Progressive Electricity Billing

Problem Statement

Calculate a household electricity bill using progressive consumption slabs, fixed charges, a surcharge, and a rebate. Compare bills for multiple months and flag an unusual increase.

Objectives

• Calculate energy charges slab by slab.

• Add fixed charges and surcharge, then apply a rebate.

• Compare monthly bills and identify a large increase.

Assumptions

• Slab rates: first 100 units ■1.50/unit; next 100 units ■2.50/unit; above 200 units ■4/unit.

• Fixed charge is ■50; surcharge is 5% of energy charges; rebate is ■20 when consumption is below 100 units.

• An increase greater than 30% from the previous month's bill is unusual.

Algorithm

1. Read the number of months.

2. For each month, read units consumed.

3. Calculate charges for each applicable slab.

4. Add fixed charge and surcharge; apply rebate if eligible.

5. Display the bill and compare it with the previous bill.

6. Flag increases greater than 30%.

Flowchart

```
START → Read months → Read units → Calculate slab charges → Add fixed fee and surcharge → Apply rebate → Comp
```

Pseudocode

```
BEGIN
  READ months
  previous_bill = 0
  FOR each month
    READ units
    Calculate energy charge using progressive slabs
    bill = energy charge + 50 + 5% of energy charge
    IF units < 100 THEN bill = bill - 20
    DISPLAY bill
    IF not first month AND bill > previous_bill * 1.30
       DISPLAY unusual increase
    END IF
    previous_bill = bill
  END FOR
END
```

Python Source Code

```
months = int(input("Enter number of months: "))
previous_bill = 0

for month in range(1, months + 1):
    units = float(input("Enter units for month " + str(month) + ": "))

    if units <= 100:
        energy_charge = units * 1.50
    elif units <= 200:
        energy_charge = 100 * 1.50 + (units - 100) * 2.50
```

```
    else:
        energy_charge = (100 * 1.50 + 100 * 2.50
                         + (units - 200) * 4.00)

    surcharge = energy_charge * 0.05
    bill = energy_charge + 50 + surcharge

    if units < 100:
        bill = bill - 20

    print("Month", month, "Bill: Rs.", round(bill, 2))

    if month > 1 and bill > previous_bill * 1.30:
        print("Warning: Unusual increase in bill!")

    previous_bill = bill
```

Test Case

Enter 2 months. Month 1: 80 units. Month 2: 180 units.
Expected bills: Month 1 = ■156.00; Month 2 = ■417.50. The second month is flagged as an unusual increase.

Sample Output

```
Month 1 Bill: Rs. 156.0
Month 2 Bill: Rs. 417.5
Warning: Unusual increase in bill!
```

Conclusion

The program applies progressive electricity rates, fixed charges, surcharge, and rebate. It also compares bills to identify a significant month-to-month increase.